

ELKINS 21 ENE, WV, USA

WMO: 724370

Lat: 39.0130N

Lon: 79.4743W

Elev: 3390

StdP: 12.98

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	3.7	-9.8	3.7	0.6	-5.4	4.7	5.7	14.0	19.1	12.8	20.6	5.3	N/A	0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	78.4	66.6	76.8	65.4	74.9	64.4	69.3	75.8	68.1	74.0	66.8	72.2	4.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	114.1	72.4	66.1	109.1	71.2	64.9	104.8	70.1	35.6	76.2	34.5	73.6	33.4	72.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	10.0	8.7	DB	-5.5	82.1	5.2	1.9	-9.3	83.5	-12.3	84.6	-15.3	85.7	-19.1	87.1	
			WB	-6.1	71.2	6.4	0.7	-10.7	71.7	-14.4	72.2	-18.0	72.6	-22.6	73.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	25.8	27.7	35.9	47.3	56.1	62.1	65.2	64.2	59.3	49.2	38.7	31.0
	DBStd	16.90	13.52	12.10	11.82	10.39	8.27	5.38	4.54	4.57	6.39	9.45	10.17	11.14
	HDD50	3140	754	628	459	170	41	0	0	0	7	131	357	593
	HDD65	6748	1215	1044	903	534	286	116	55	70	190	491	790	1054
	CDD50	2039	4	4	22	89	230	364	470	440	286	108	17	5
	CDD65	167	0	0	0	3	9	29	60	45	18	3	0	0
	CDH74	556	0	0	0	16	39	89	187	140	75	10	0	0
	CDH80	18	0	0	0	0	0	3	11	2	2	0	0	0
Wind	WSAvg	3.9	5.2	5.3	4.8	4.8	3.7	3.2	2.7	2.5	2.7	3.5	4.0	4.8
Precipitation	PrecAvg	44.6	3.5	3.2	4.1	3.8	4.5	4.3	4.7	3.9	3.4	3.0	2.6	3.6
	PrecMax	65.4	6.5	6.8	7.3	7.8	9.4	7.2	8.8	7.7	8.8	5.1	5.6	6.3
	PrecMin	35.2	1.8	1.2	1.4	1.7	1.7	1.3	2.1	2.2	0.4	0.7	0.5	1.5
	PrecStd	6.8	1.1	1.3	1.5	1.5	1.9	1.6	1.8	1.4	2.1	1.5	1.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	60.6	67.7	76.4	77.8	79.3	81.6	80.1	79.2	75.3	66.9	59.6
		MCWB	52.1	53.4	52.6	55.3	62.6	67.1	70.6	68.7	66.0	62.2	51.8	53.6
	2%	DB	54.3	54.3	62.1	71.4	74.8	76.7	78.3	77.7	76.3	70.6	60.8	55.1
		MCWB	49.4	49.3	51.4	54.4	61.0	65.2	68.3	66.7	64.4	59.2	50.6	50.4
	5%	DB	50.4	50.4	57.6	67.4	72.0	74.3	76.0	75.4	73.4	67.1	56.8	51.4
		MCWB	45.2	45.7	48.7	53.2	60.6	64.1	67.0	65.4	63.7	58.0	48.2	47.7
	10%	DB	46.2	46.0	53.7	63.1	68.9	71.5	73.8	73.1	70.2	63.7	53.4	48.0
		MCWB	42.4	40.4	46.9	51.1	58.9	63.2	65.5	64.4	62.9	56.2	46.7	44.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	55.3	56.0	59.9	66.6	68.8	72.5	69.7	69.0	65.4	56.7	55.0
		MCDB	57.2	61.1	64.4	70.4	74.1	76.2	78.0	76.2	75.4	72.5	63.9	58.5
	2%	WB	50.4	52.3	53.2	57.1	64.2	67.0	70.1	68.3	67.3	63.4	53.4	52.4
		MCDB	53.9	55.8	58.9	65.9	71.5	73.4	76.9	74.1	73.5	68.3	58.3	55.4
	5%	WB	46.3	48.9	51.0	55.2	62.8	65.7	68.6	67.2	65.9	61.1	50.5	49.7
		MCDB	49.0	51.8	56.4	62.5	69.6	71.8	74.6	72.3	71.1	65.6	54.2	52.2
	10%	WB	41.8	44.2	48.7	53.1	61.3	64.5	67.4	65.9	64.2	58.0	47.5	46.9
		MCDB	44.0	47.6	53.5	60.1	67.4	69.8	72.6	70.7	68.8	62.4	51.6	49.6
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.0	16.0	18.5	16.6	15.6	15.6	16.2	17.6	16.7	15.7	13.4
		MCDBR	15.7	19.0	20.5	22.8	20.2	18.1	17.9	18.3	20.7	20.9	19.7	17.5
		MCWBR	13.3	14.3	12.4	11.1	9.0	9.4	8.6	9.3	9.7	10.5	12.3	12.8
	5% WB	MCDBR	16.4	18.8	17.7	17.5	16.3	15.1	15.5	14.7	15.8	15.9	16.4	15.8
		MCWBR	14.2	15.4	12.2	10.2	8.3	8.4	8.1	7.9	8.4	8.6	12.5	13.3
Clear-Sky Solar Irradiance	taub	0.270	0.286	0.310	0.351	0.384	0.419	0.433	0.443	0.358	0.319	0.287	0.270	
	taud	2.524	2.460	2.468	2.382	2.286	2.212	2.198	2.140	2.378	2.502	2.564	2.557	
	Ebn at Noon	292	300	302	293	284	273	268	262	281	284	284	285	
	Edh at Noon	23	29	32	37	42	45	45	47	35	28	23	21	
All-Sky Solar Radiation	RadAvg	575	833	1151	1495	1655	1832	1778	1622	1352	991	700	507	
	RadStd	60	122	82	126	146	129	108	85	163	117	75	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+110

Nomenclature: See separate page